

PRINCE MODI

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EDUCATION

University of California, San Diego

Expected Mar 2027

MS Computer Science | GPA: 3.88/4.00

Ganpat University, India

Jul 2022

B. Tech. Computer Engineering | GPA: 3.96/4.00 | Academic scholarship; ranked 2nd of 60+

EXPERIENCE

Picasso Lab, UC San Diego

Jun 2026 - Present

Graduate Student Researcher, LLM Inference Systems

- Developing a **trace-driven simulator** to select context, tensor, data, and expert parallelism for agentic LLM prefill and append-prefill workloads using measured attention costs.
- Investigating collective-communication synchronization overhead in attention and mixture-of-experts layers.

Indian Institute of Science (IISc), Bangalore

Jan 2023 - Nov 2024

Project Associate, Distributed Systems

Full-time through Apr 2024; then volunteer research collaborator

- Built **Flotilla**, a Python/AsyncIO federated learning framework for **1,000+ workers**, supporting synchronous and asynchronous aggregation through a pluggable event-driven state machine.
- Designed a **Redis-backed parameter server** with stateless workers, reducing per-round overhead to **1.7%** at 1,000+ nodes versus **54.9%** for an open-source baseline.
- Implemented **gRPC/MQTT** worker discovery, heartbeat monitoring, incremental checkpoints, and hot failover; recovered a live session on a standby server within **820 ms**.
- Validated CNN, LSTM, and ALBERT training across NVIDIA Jetson GPUs and ARM devices, tolerating **~40% worker dropout** with negligible accuracy impact.

Sterlite Technologies Limited, Ahmedabad

Jan 2022 - Aug 2022

Software Engineering Intern

- Refactored Docker images in Linux CI/CD pipelines, reducing image size by **35%** and build time by **50%**.
- Integrated Liquibase to track MongoDB schema changes across environments.

SELECTED PROJECTS

Distributed Container Snapshotting | Code

Jun 2026

- Built **Chandy-Lamport snapshots** of live containers with Python, CRIU, and Podman, plus a transparent reliable-UDP sidecar with sequencing and retransmission.
- Verified consistent counter state after multi-host snapshot/restore under concurrent traffic, delay, and packet loss.

GPU Kernel Performance and Portability | Code

Feb 2026

- Implemented **FlashAttention-2** and **RMSNorm** in Helion, Triton, Gluon, and CuTe; benchmarked throughput and utilization across **four NVIDIA GPU generations**.
- Used Nsight Compute to identify memory-throughput losses on T4 for A100-tuned Triton/Gluon kernels.

SELECTED PUBLICATIONS & POSTERS

NeurIPS 2026 InterpScience workshop (poster): Autointerp Simulation Scores Track Token Naming, Not Explanation Truth.

SOSP 2026 (poster): Recovering Internal Command Interfaces from Applications for Direct Agent Invocation.

JPDC 2025 (journal): Flotilla: A Scalable, Modular and Resilient Federated Learning Framework for Heterogeneous Resources.

TECHNICAL SKILLS

Languages: Python, C, Go, Java, Bash/Shell

ML systems: PyTorch, Hugging Face, SGLang, NCCL, Triton, Nsight Compute

Infrastructure: Linux, Docker/Podman, CRIU, CI/CD, AWS EC2, gRPC, MQTT, Redis, PostgreSQL, Git